

Business Problem Statement

A retail company wants to understand its Diwali Sales Trends in order to improve sales, customer satisfaction and profit. The management team has noticed changes in purchasing patterns among Gender, Age group and location .They are interested in uncovering which factors, such as age, gender, occupation and people from different location drive consumer decisions for purchasing different items . You are tasked with analyzing the company's Diwali Sales dataset to answer the following business question:

“How can the company leverage consumer Diwali Sales data to identify trends, improve customer engagement, and optimize marketing and product strategies?”

Deliverables

1. Data Preparation & Modeling (Python): Clean and transform the raw dataset for analysis.
2. Data Analysis and Visualization (Python, Seaborn): Used exploratory data analysis (EDA) with Seaborn for analyzing the gender, state, age, marital status count and respective sum of amounts and format them into graphs.
3. Report and Presentation: Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.
4. GitHub Repository: Include all Python scripts, and dashboard files in a well-structured repository.